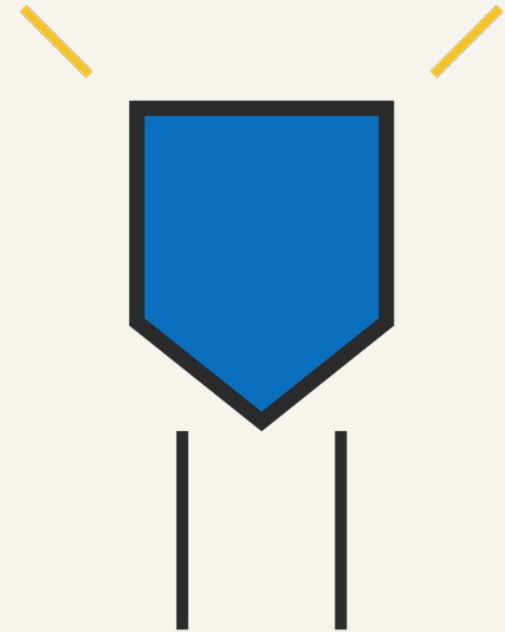


Semiconductors & p-n Junctions

Why some solids conduct and others don't.

Higher Physics



Starter

2009 Higher Physics

Scan the code and answer these:

Q8

Q11

Section A · charged particles, capacitors

Show your working, not just the letter.



**Scan for the paper
Marking scheme**

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PART 1

Energy bands

Every solid has its own band structure — that structure is what decides whether it conducts.

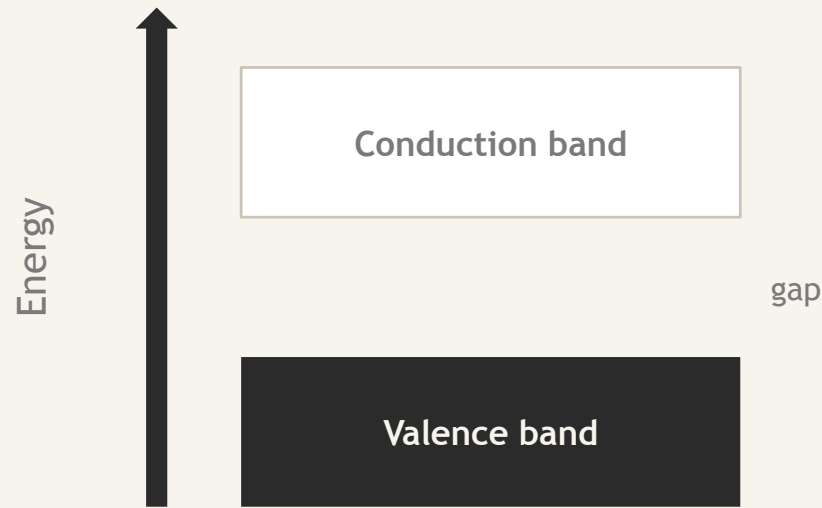
By the end of this part...

- 1 I can describe how energy bands form from the energy levels of individual atoms
- 2 I can name and describe the conduction band and the valence band
- 3 I can use band structure to explain why a material is a conductor, an insulator or a semiconductor

From energy levels to energy bands

- In a single atom, electrons sit in discrete energy levels.
- In a solid, close atoms merge those levels into energy bands.
- A gap between bands is a range of energies no electron can have.

Conduction band and valence band



Valence band — contains the electrons that bond the solid together.

Conduction band — the band an electron must reach to move freely and carry a current.

Conductors



Conductor

- The bands overlap, or the conduction band is already partially filled.
- Free electrons and empty states are both available — no gap to cross.
- Result: highly conductive — a metal.

Insulators

Conduction band

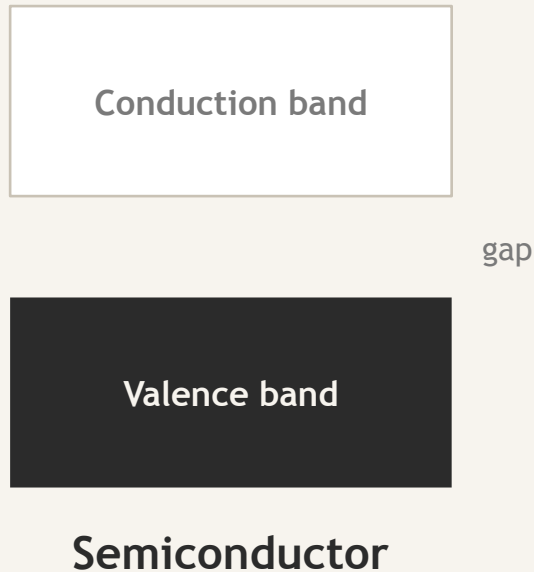
gap

Valence band

Insulator

- The valence band is full and the gap to the conduction band is large.
- Not enough energy to cross the gap at room temperature.
- Result: no conduction.

Semiconductors



- The valence band is full, but the gap is smaller than in an insulator.
- At room temperature, some electrons have enough energy to cross the gap.
- Result: some conduction.
- Conductivity increases with temperature as more electrons cross.

Your turn – classify it

SHOW ME BOARDS

- 1 A material's valence and conduction bands overlap. What kind of material is it?
- 2 A material has a full valence band and a very large energy gap. What kind of material is it, and why does it not conduct?
- 3 Explain why a semiconductor conducts better when it is warmed up.

TRY THIS – Answers

- 1 A conductor – free electrons and empty states are available with no gap to cross.
- 2 An insulator – the gap is too large to cross at room temperature, so the conduction band stays empty.
- 3 Raising the temperature gives more electrons enough energy to cross the small gap – more charge carriers become available.



Conduction band and valence band

Conduction band — the band an electron must reach to move freely and carry a current.

Valence band — the band containing the bonding electrons.



Conductors, insulators and semiconductors

Conductor: bands overlap / conduction band partly filled — highly conductive.

Insulator: valence band full, large gap — no conduction.

Semiconductor: valence band full, small gap — some conduction, increasing with temperature.

Task pointer

Open your booklet at Section 1.
Complete the band-structure questions.

Booklet p.1

Starter

2013 Higher Physics

Scan the code and answer these:

Q8

Q11

Section A · charged particles, capacitors

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PART 2

Doping and the p-n junction

Add the right impurities to a semiconductor, and you can control exactly how it conducts.

By the end of this part...

- 1 I can describe doping and explain why it is used
- 2 I can distinguish p-type from n-type semiconductors
- 3 I can describe what happens when p-type and n-type layers meet

Doping

- During manufacture, a semiconductor can be doped — small amounts of a specific impurity are added.
- Doping is done deliberately to increase conductivity in a controlled way.
- Doping produces two types of semiconductor: p-type and n-type.

p-type and n-type

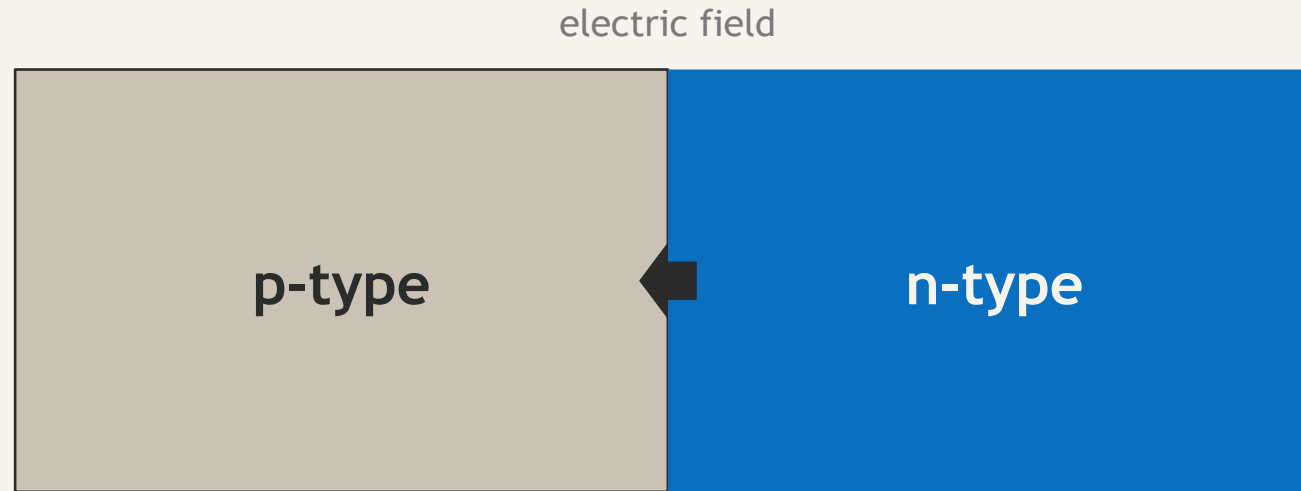
p-TYPE

- Majority charge carriers are positive
- ("p" for positive)

n-TYPE

- Majority charge carriers are negative / electrons
- ("n" for negative)

The p-n junction



When a semiconductor has p-type and n-type doping in adjacent layers, a p-n junction is formed.

There is an electric field in the p-n junction.

Your turn – doping and junctions

SHOW ME BOARDS

- 1 What is meant by doping a semiconductor?
- 2 A semiconductor's majority charge carriers are electrons. Is it p-type or n-type?
- 3 What is produced when p-type and n-type layers are next to each other, and what does it contain?

TRY THIS – Answers

- 1 Adding specific impurities to a semiconductor, during manufacture, to increase its conductivity in a controlled way.
- 2 n-type – the majority carriers are electrons, which are negative.
- 3 A p-n junction – it contains an electric field.



Doping and the p-n junction

Doping — adding specific impurities to a semiconductor to increase its conductivity.

p-type — majority carriers positive. n-type — majority carriers negative / electrons.

p-n junction — formed where p-type and n-type layers are adjacent; contains an electric field.

Task pointer

Open your booklet at Section 2.
Complete the doping and p-n junction
questions.

Booklet p.2

Starter

2015 Revised Higher Physics

Scan the code and answer these:

Q17

Q19

Q20

Section A · a.c., semiconductors

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PART 3

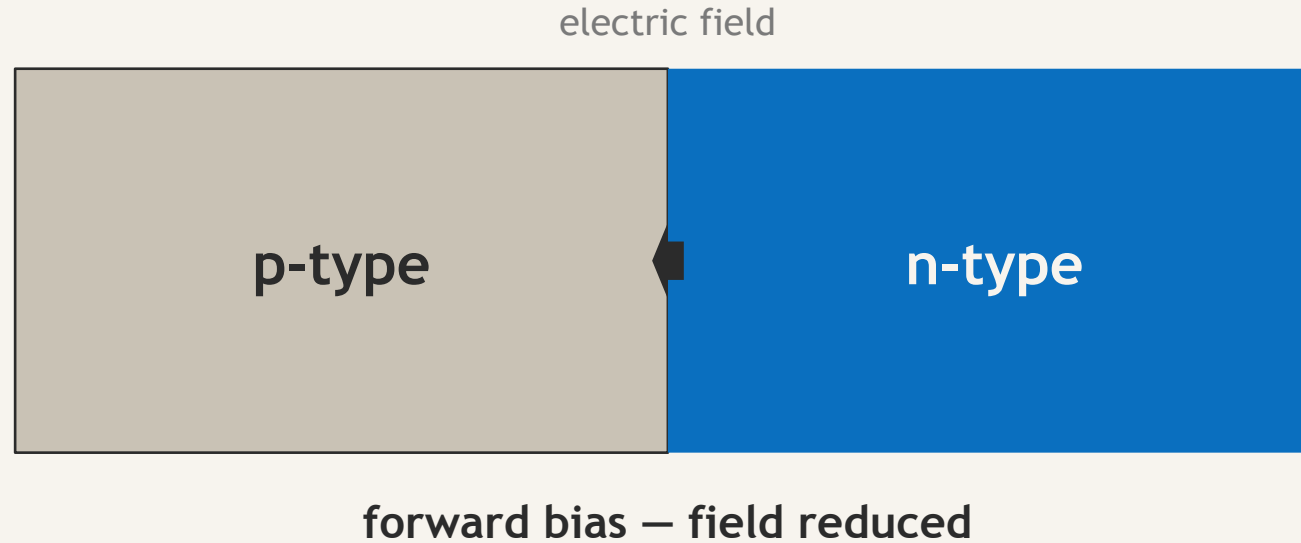
Forward and reverse bias

Connect a p-n junction one way, and the field shrinks. Connect it the other way, and it grows.

By the end of this part...

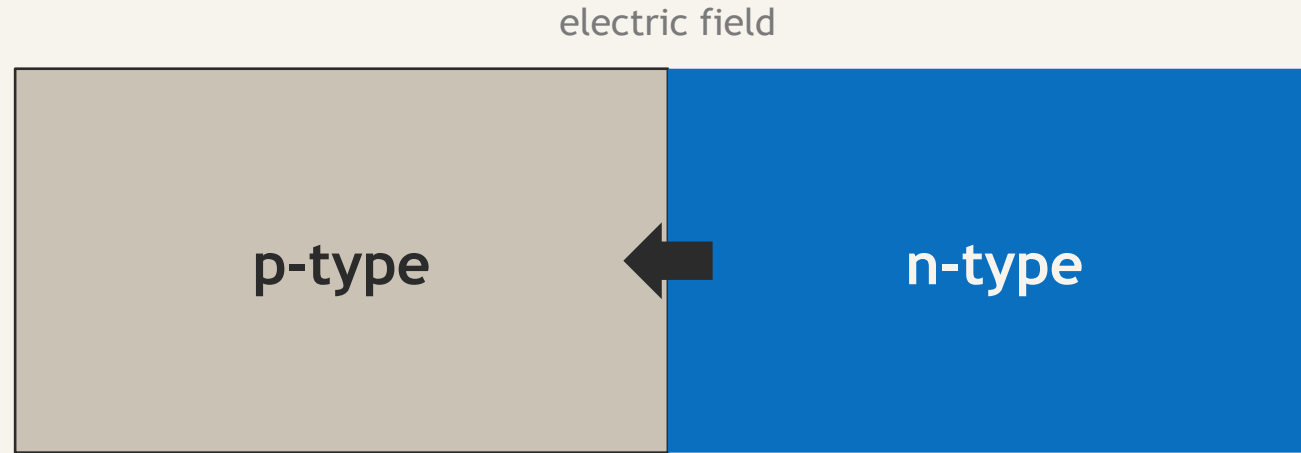
- 1 I can state what forward bias and reverse bias mean
- 2 I can state the effect of each on the electric field in the junction

Forward bias



- Forward bias connects the supply so the electric field in the junction is reduced.

Reverse bias



reverse bias — field increased

- Reverse bias connects the supply so the electric field in the junction is increased.

Watch Out – say which bias, precisely

THE MISTAKE

- "This LED conducts because it's biased."
- Not naming which bias lets current flow

THE CORRECT ANSWER

- Name the bias: forward or reverse.
- Forward reduces the field; reverse increases it.



Forward and reverse bias

Forward bias — reduces the electric field in the p-n junction.

Reverse bias — increases the electric field in the p-n junction.

Task pointer

Open your booklet at Section 3.
Complete the bias questions.

Booklet p.3

Starter

2014 Higher Physics

Scan the code and answer these:

Q10

Q11

Section A · a.c., capacitors

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PART 4

Light-emitting diodes

An LED is a forward-biased p-n junction doing one very specific job: turning electrons into photons.

By the end of this part...

- 1 I can describe an LED as a forward-biased p-n junction diode
- 2 I can describe, in terms of bands, how an LED emits a photon

What an LED is

- An LED is a forward-biased p-n junction diode that emits photons.
- Forward bias pushes electrons from the n-type conduction band towards the p-type conduction band.

How the photon is emitted



An electron falls from the conduction band into the valence band, either side of the junction.

A photon is emitted. Its energy equals the band gap.

Watch Out – the band gap sets the colour

THE MISTAKE

- Explaining LED colour without mentioning the band gap
- Treating every LED's photon energy as the same

THE CORRECT ANSWER

- Photon energy equals the band gap.
- Bigger gap: higher-energy, bluer photon.
- Smaller gap: lower-energy, redder photon.



Light-emitting diodes (LEDs)

An LED is a forward-biased p-n junction diode that emits photons.

Forward bias moves electrons from the n-type conduction band towards the p-type conduction band.

Photons are emitted when electrons fall from the conduction band into the valence band, either side of the junction.

The photon energy equals the band gap.

Task pointer

Open your booklet at Section 4.
Complete the LED questions.

Booklet p.4

Starter

2014 Revised Higher Physics

Scan the code and answer these:

Q8

Q16

Q18

Section A · particles, semiconductors

Show your working, not just the letter.



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PART 5

Solar cells

Run the LED process backwards, and light goes in instead of coming out — that is a solar cell.

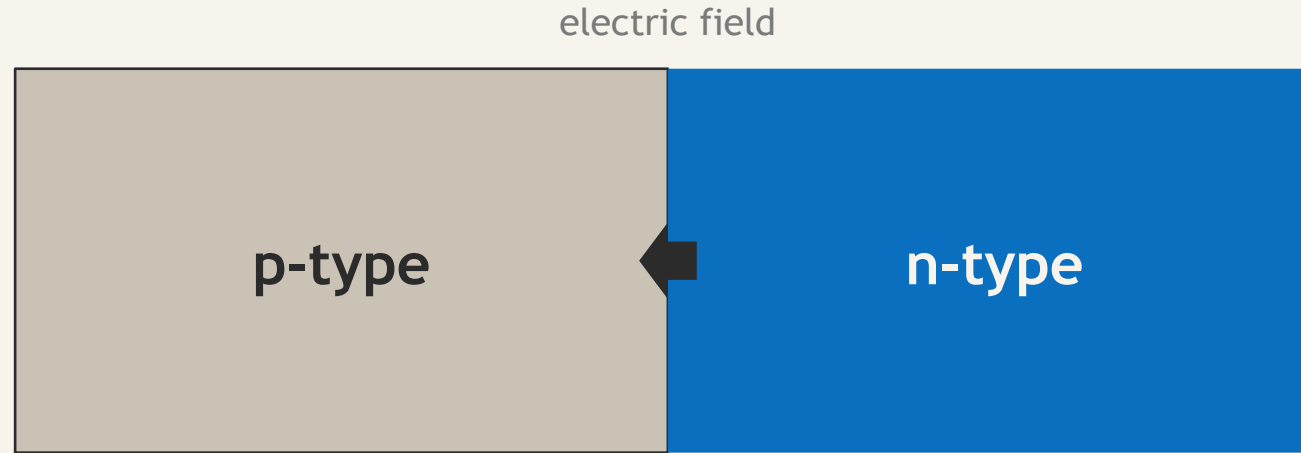
By the end of this part...

- 1 I can describe a solar cell as a p-n junction and name the photovoltaic effect
- 2 I can describe, in terms of bands, how a solar cell produces a p.d.

The photovoltaic effect

- A solar cell is a p-n junction designed to produce a potential difference when photons are absorbed.
- This is the photovoltaic effect.
- A photon is absorbed, giving an electron enough energy to be raised into the conduction band.

How the p.d. is produced



The p-n junction's electric field causes the electron, once raised to the conduction band, to move towards the n-type side. This produces a potential difference across the solar cell.

Watch Out — solar cell, not LED

THE MISTAKE

- Describing a solar cell as if it emits photons, like an LED
- Mixing up which device absorbs and which one emits

THE CORRECT ANSWER

- A solar cell absorbs a photon; an LED emits one.
- Always check which device the question names.



Solar cells

A solar cell is a p-n junction designed to produce a p.d. when photons are absorbed – the photovoltaic effect.

A photon's energy raises an electron from the valence band to the conduction band.

The junction's field moves that electron towards the n-type side, producing a p.d. across the cell.

Your turn – solar cells

SHOW ME BOARDS

- 1 Name the effect by which a solar cell produces a potential difference.
- 2 What happens to an electron when a solar cell absorbs a photon?
- 3 A solar cell and an LED are both p-n junctions. What is the key difference in how each one is used?

TRY THIS – Answers

- 1 The photovoltaic effect.
- 2 It is raised from the valence band to the conduction band.
- 3 A solar cell absorbs photons to produce a p.d.; a forward-biased LED emits them – opposite directions.

Checkpoint

- 1 State the difference between the valence band and the conduction band.
- 2 A material conducts well and its conductivity barely changes with temperature. What kind of material is it, and why?
- 3 Explain, in terms of band structure, why a semiconductor's conductivity increases with temperature.

Exam Practice

2011 Q18 – n-type majority carriers.
2014 Q18 – doping a pure semiconductor.
2015 (Revised) Q19 – energy bands.

Past papers, Section A

LOOKING AHEAD

Next: Motion, Graphs & Projectiles

Vectors, scalars, and the equations that connect them — the largest deck of the course.

Plenary

- 1 Name the two bands every solid's electrons occupy.
- 2 Explain, using band structure, why an insulator does not conduct.
- 3 What is doping, and why is a semiconductor doped?
- 4 State the effect of forward bias on the electric field in a p-n junction.
- 5 An LED emits light and a solar cell absorbs it. Explain the difference in terms of the electron's journey between bands.

- 1 Two LEDs emit different colours. Which one has the larger band gap?
- 2 Why does reference to Fermi levels not appear anywhere in this topic's answers?

- 1 A pure semiconductor is doped n-type. Sketch its band diagram before and after, and explain what changed.
- 2 Solar panels lose efficiency as they heat up, even though semiconductor conductivity rises with temperature. Suggest why.

Extension — EXTRA

EXTRA PRACTICE

- 1 Research one real semiconductor material used in LEDs other than silicon, and find its approximate band gap.